

PROJECT TECHNICAL REPORT
National Level Submission — Arduino Physical AI Challenge India 2026

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 1. PROJECT ARCHITECTURE PROFILE

Developer : Rahul Kalwa
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Competition : Arduino Physical AI Challenge India 2026
Track : Industrial, Smart Manufacturing & Sustainability

 2. DEVELOPER & INSTITUTIONAL PROFILE

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COUNTRY : India
SUBMISSION ID : APC-2026-RJ-50225

 3. INTRODUCTION & VISION

Traditional power distribution networks face massive challenges worldwide, including delayed fault detection, manual diagnostic latency, transformer core overheating, and cascading grid failures that lead to widespread regional blackouts.

The vision of this project is to architect a completely self-reliant, resilient, and intelligent 'Smart Grid' using cutting-edge embedded chipsets, synchronized sensor arrays, and localized Edge AI inference processing. Driven by the high-speed processing capabilities of the Arduino® UNO™ Q, this system actively identifies transmission faults, estimates their exact structural distance, isolates zonal errors, and switches onto auxiliary backup micro-grids in real-time, effectively keeping manual human grid intervention to an absolute minimum.

 4. SYSTEM HARDWARE INFRASTRUCTURE & BOM

The implementation relies on a deeply integrated multi-layer embedded architecture utilizing the following hardware framework:

1. PRIMARY EDGE PROCESSING UNIT:

* Arduino® UNO™ Q (The foundation micro-computing board used for multi-channel sensor telemetry sampling, mathematical line resistance calculation, and rapid fault trip execution).

2. IOT GATEWAY & TELEMETRY MODULE:

* NodeMCU ESP8266 (Dedicated processing unit configured for processing TCP/IP protocols, uploading operational metrics to the remote cloud server, and sending emergency warnings).

3. SENSOR NETWORK MATRIX:

* ACS712 Hall-Effect Isolated Current Sensors (For continuous AC load and overload monitoring).

* High-Resolution Linear Thermal Sensors (For tracking the transformer's core operational heat).

* Transistorized High-Sensitivity Wireless Electromagnetic Phase Detectors.

4. POWER SWITCHING & SWITCHGEAR DRIVERS:

* Multi-Channel Opto-isolated Relays, a heavy-duty 12-0-12V 5A step-down transformer assembly, and a simulated 200W automated changeover emergency inverter sub-grid.

5. CORE ARCHITECTURE & PHASE-BY-PHASE FUNCTIONAL WORKING

PHASE 1: 33KV INCOMING GRID & HIGH-SPEED FAULT ISOLATION

* OBJECTIVE: To provide instantaneous switchgear protection to high-voltage incoming transmission

lines right at the entry point of the substation.

* WORKING METHODOLOGY: This section manages the simulated high-tension 33KV input via dual

interlocked relay channels and an integrated optical fire safety module. An active Ground (GND)

trace plane is physically mapped directly beneath and between the three dummy transmission lines—

representing Phases R, Y, and B.

* EDGE RESPONSE: The instant a phase line snaps or encounters a physical ground short-circuit

(Phase-to-Ground Fault), a huge current surge is registered. The Arduino control loop detects this anomaly and fires a trip pulse to the isolation relays within less than 1 millisecond. This stops downstream electrical fires and protects the main substation framework from permanent failure.

PHASE 2: 11KV DISTRIBUTION TRANSMISSION & DISTANCE ESTIMATION ANALYSIS

- * **OBJECTIVE:** To instantly detect phase short-circuits across long-distance distribution lines and calculate the precise distance of the fault from the control office.
- * **WORKING METHODOLOGY:** The 11KV transmission line grid is physically modeled through a series of
 - 5 dummy towers. Precise, heavy-duty resistors are integrated into the sequence between each tower to mimic natural, distance-dependent Line Resistance (R_{line}).
- * **EDGE RESPONSE:** When a localized short circuit occurs at any tower point along the lines, the
 - total circuit resistance drops in a linear, predictable pattern. The local edge-inference algorithm on the Arduino evaluates the drop in the voltage divider network and locks the precise, real-time fault characteristics onto the central station alphanumeric LCD:
 - **FAULT DIRECTION/TYPE:** Identifies the specific phase under error (R, Y, or B).
 - **BREAKDOWN CURRENT:** Captures the absolute peak short-circuit tripping current (A).
 - **EXACT FAULT DISTANCE:** Solves the resistance equation and outputs the distance in kilometers (e.g., 2.0 KM or 4.0 KM).
- * **DATA LOCK MECHANISM:** For strict field safety and maintenance logging, this diagnostic telemetry
 - remains securely locked on the display screen and cannot be cleared until an authorized technician manually clears the grid lines and presses the 'Physical Reset' button.



PHASE 3: WIRELESS UNDERGROUND CABLE TRACKER (ROBOTIC EDGE DETECTION)

- * **OBJECTIVE:** To pinpoint underground cable snaps or structural breakdowns without requiring costly
 - and blind continuous street excavation.
- * **WORKING METHODOLOGY:** A model underground pipeline channel mimics a buried power transit route
 - where a section of the electrical conductor is broken (Fault Zone). An autonomous robotic tracking rover is engineered to drive over this path.
- * **EDGE RESPONSE:** The tracking rover houses an onboard, highly sensitive multi-stage transistor
 - circuit tuned to detect close-range electromagnetic waves. As long as the rover drives over an intact, live underground cable, it continuously captures the magnetic flux field, keeping its status LED lit. The moment it passes over the broken section (Fault Zone), the electromagnetic field instantly disappears and the LED shuts off. An onboard Light Dependent Resistor (LDR) logs this state transition immediately, prompting the rover to stop and broadcast its exact coordinates

back to the substation.

PHASE 4: TRANSFORMER THERMAL SAFETY & TWO-STAGE AUTOMATION PROTECTION

* OBJECTIVE: To eliminate distribution transformer burnouts caused by long-term overload states or

cooling oil breakdown.

* DUAL-LAYER AUTOMATION LOGIC: Thermal sensors constantly stream raw core heat metrics back to the

main processor, which runs a continuous two-stage protective algorithm:

- STAGE 1 (ACTIVE FORCED COOLING): If the core temperature rises past the safe operational limit

due to heavy consumer demand, the processor triggers a high-velocity cooling fan array to manage

the heat without cutting consumer grid power.

- STAGE 2 (EMERGENCY LOAD SHUTDOWN): If the temperature continues to rise to dangerous levels

despite active cooling, the controller trips the main line circuit breaker, shutting down the input feed to protect the internal copper windings from melting.

PHASE 5: AUXILIARY MICRO-GRID EMERGENCY BACKUP SYSTEM (VILLAGE UPS GATEWAY)

* OBJECTIVE: To provide an uninterrupted auxiliary electricity feed to critical regional zones during structural 11KV line failures.

* WORKING METHODOLOGY: This system integrates a 200W step-up power inverter network paired with a

smart charging controller and dual automatic changeover high-current relay circuits.

* EDGE RESPONSE: When a major storm or an unexpected failure cuts off power from the primary 11KV

grid, the Arduino unit senses the zero-voltage state and switches lines smoothly. It safely shifts critical local lines over to the auxiliary backup inverter network. This minimizes sudden blackouts

for consumers and allows repair crews to fix the heavy 11KV transmission lines under zero-pressure, unenergized conditions.

PHASE 6: ZONAL ISOLATION CONTROL (PHASE-ISOLATION INTERACTION)

* OBJECTIVE: To confine local line faults to a single street or community zone, preventing an error

in one area from causing a complete blackout across the entire grid.

- * **WORKING METHODOLOGY:** The three phases coming from the distribution transformer secondary windings are split into separate residential grid pathways (Zone 1 connected to Phase R, Zone 2 to Phase Y, and Zone 3 to Phase B).
- * **EDGE RESPONSE:** If a localized line short-circuit occurs strictly within Zone 1 (R-Phase), the controller isolates and opens the circuit breaker for Zone 1 only. Phases Y and B remain live and continue delivering clean, stable power to Zones 2 and 3, reducing unnecessary community blackouts by 66%.

PHASE 7: SMART BALANCE PROTECTION (PHASE-LOAD BALANCING VIA ARDUINO UNO Q EDGE AI)

- * **OBJECTIVE:** To protect distribution infrastructure from phase-unbalance and localized core damage when sudden consumer consumption spikes occur on a single phase line.
- * **WORKING METHODOLOGY:** ACS712 sensors sample current telemetry data at high frequencies from all three phase lines simultaneously.
- * **EDGE RESPONSE:** If consumer behavior in any particular zone causes power draw to cross the dynamically allocated safety envelope, the ACS712 registers an overcurrent event. Rather than shutting down the main transformer and affecting the entire town, the controller acts quickly to trip just the overloaded relay pathway, protecting the core asset until load equilibrium is restored.

PHASE 8: COMPREHENSIVE SMART CONSUMER METER & HOME APPLIANCE GATEWAY

- * **OBJECTIVE:** To enforce accurate energy monitoring at the consumer end, prevent data tampering, and protect expensive home appliances from line voltage anomalies.
- * **WORKING METHODOLOGY:** Installed right at the residential power entry hub, this digital meter runs anti-tamper firmware and executes a strict, three-layer hardware safety sequence:
 - **UNDER-VOLTAGE ALERT:** If grid brownouts cause voltage to fall below safe operating limits, the meter prints a clear error message on the screen: `[Voltage Down! Reduce Load!]` to warn users to turn off heavy inductive loads manually.

- OVER-VOLTAGE ISOLATION: In the event of a sudden, dangerous high-voltage line spike, the onboard relay isolates the internal home circuitry in under 1 millisecond, keeping high-end electronics (TVs, Refrigerators, AC units) safe from power surges.
- 3KW DOMESTIC CAPACITY TRIP: The firmware enforces a maximum power consumption cap of 3kW. If the consumer turns on excess equipment that oversteps this limit, the meter locks down the home line and sends a capacity breach alert directly back to the central distribution office.

6. CONCLUSION & FUTURE DEVELOPMENT SCOPE

The AI-Driven Smart Grid Power Distribution System successfully demonstrates an automated, self-healing infrastructure model tailored for modern electrical grids. By running real-time edge calculations on the Arduino® UNO™ Q and integrating smart protective switchgear, the design avoids the delays of manual inspection and maintenance.

Future phases will focus on deploying wide-area wireless mesh networks and incorporating machine learning models for predictive maintenance analytics, aiming to make rural distribution grids virtually fail-safe.

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THANKS FOR WATCHING MY PROJECT
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